

Designing Better Transformers: Using Amorphous Network Proxies to Engineer the Unified Highway Transformer

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Abstract: Standard transformer architectures define information flow through rigid, serialized sequences of discrete operators. In deep or very deep implementations, this leads to PreNorm Dilution, or Latent Amnesia, in which deep layers become blind to early layer signals in the residual stream, stalling convergence. Existing solutions to PreNorm Dilution are computationally expensive or add engineering complications. This work utilizes unconstrained Amorphous Neural Network (AmNN) clouds as diagnostic proxies allowing analysis of information flow that emerges free from the constraints of discrete operators. These paths empirically confirmed the demand for historical latent context in deeper layers. Modeling this flow back into a discrete operator transformer resulted in the creation of the Unified Highway Transformer (UHT). The UHT introduces an $O(1)$ parallel memory bus (Z_{mem}) that builds a history of latent context from the residual stream and makes that history available to deep layers. To counter initialization instability during training, a Dynamic Variance Annealer is introduced, to initially scale residual projections entering the $O(1)$ bus, phasing out that scaling completely after initial epochs. The UHT was evaluated against a constrained 120-layer, 9.7-million total trainable parameter architectural stress test ($d_{model} = 64$) on WikiText-103. A gradient tracker revealed dynamic, high-variance, gradient norm activation at deep levels in the UHT, illustrating that the UHT restored the representational capacity of the deep layers. In contrast a 120-layer Pre-LN control showed gradient stagnation in deep layers. The UHT resolved the Latent Amnesia and utilized the full depth of the deep network. The UHT reduced terminal perplexity in the tested configuration by more than half (169 to 74), maximizing the empirical capacity of this severely bottlenecked model. An 180-layer test of the UHT showed similar performance, empirically supporting the conclusion that the constrained model ($d_{model} = 64$) reached theoretical limits of performance on WikiText-103. The UHT provides a scalable, parameter-efficient architecture for extreme-depth modeling that solves gradient stagnation in deep networks.

1. Introduction

Transformers, the backbone of deep learning for language models, rely on sequences of discrete operators that inherently create a rigid information flow. This flow is strictly serialized, passing through specialized modules—such as Multi-Head Self-Attention (MHSA) and Feed-Forward Networks (FFNs)—which exchange data with a primary residual stream.

Recent literature discussing Latent Amnesia (Manson, 2025) describes it as a phenomenon where deep networks progressively lose access to their own early-layer representations. Earlier this year, Moonshot AI referenced the mathematical root of this pathology as 'PreNorm Dilution' (Kimi Team, 2026), proving that standard unscaled residual connections physically dilute early state contributions. To solve this, Moonshot AI introduced Attention Residuals, a method that

interconnects layers with attention outputs from prior layers. That method introduces cross-layer communication bottlenecks, chunking ('Block AttnRes') to manage distributed memory transfers, and requires cached attention states to be available during inference.

This work utilized Amorphous Neural Networks (AmNN) as a diagnostic tool. In a Pre-LN Transformer, the series of discrete FFN operators and the series of self-attention operators were each replaced with an amorphous cloud that sampled data from and injected data to the residual signal stream at various points. These dual clouds created an environment where information flow could be governed by emergent dynamics within the clouds.

When tested on the WikiText dataset, this structure successfully learned. These unstructured amorphous clouds effectively replaced multiple discrete neural network operators. Analyzing the emergent flow within the amorphous clouds yielded unexpected diagnostic insights. The organic routing patterns revealed new critical pathways in this hybrid cloud transformer that also indicated inefficiencies inherent to the traditional Transformer structure. One of the major pathways the data revealed was the hybrid self-attention cloud's preference for upstream cloud data to mix with newly injected data from the residual stream. The FFN cloud showed a similar, but weaker, desire for upstream data. This is a direct confirmation that forcing the self-attention block to operate only from the normalized residual stream is a structural inefficiency in the standard Transformer.

Recognizing that today's industrial infrastructure is not ready to efficiently handle amorphous flow, the cloud flow analysis was used as a guide to make minimally invasive modifications to the baseline Transformer and improve its performance. In this process, the unstructured amorphous clouds were a powerful diagnostic tool.

This work introduces the Unified Highway Transformer (UHT). It includes a "memory stream," or "unified memory highway" (Z_{mem}), that is periodically updated via memory projections and direct addition from the residual stream. The Z_{mem} bus is implemented as an $O(1)$ parallel memory bus. The memory stream serves as an "upstream" flow for each of the discrete operators—both the self-attention and the FFN operators. It acts as a collective latent space that interconnects the discrete operators and provides context otherwise "forgotten" in a conventional Transformer. The Z_{mem} bus models the cloud behavior in the diagnostic workbench, and removes amnesia caused by sole reliance on the residual stream. These benefits are achieved without separate cross-layer communication, temporal state caching, or the distributing latent training primitives.

This work established:

A Powerful Method: AmNN proxy clouds are a diagnostic tool to map unconstrained information flow.

A Powerful Transformer Architecture: The new Unified Highway Transformer (UHT) featuring an $O(1)$ parallel memory bus (Z_{mem}) effectively learns in deep networks.

Empirical Results: In a 120-layer, 9.7M parameter intentional stress test on WikiText-103, the UHT eliminated deep-layer gradient stagnation and reduced terminal perplexity from 169 in the Pre-LN baseline to 74. A 180-Layer UHT confirmed the architecture extracts maximum theoretical capacity for a 64-dimension model.

2. Architectural Methodology – the Amorphous Diagnostic

The foundational mechanics of Amorphous Neural Networks—referred to hereafter as AmNNs—were recently formalized in literature (Simon, 2026). As established by Simon, these networks operate via unconstrained connections where a neuron's inputs are stochastically distributed to upstream neurons at multiple upstream depths. This creates a non-isomorphic, fluidic relationship between processing states, bypassing the rigid constraints of traditional discrete layers.

While this prior literature establishes the fundamental structural theory of amorphous topologies, this paper introduces their novel application as a diagnostic workbench to evaluate discrete-operator flow. Figure 1 illustrates the approach taken to develop a dual-cloud model used to diagnose transformer flow, implemented in PyTorch.

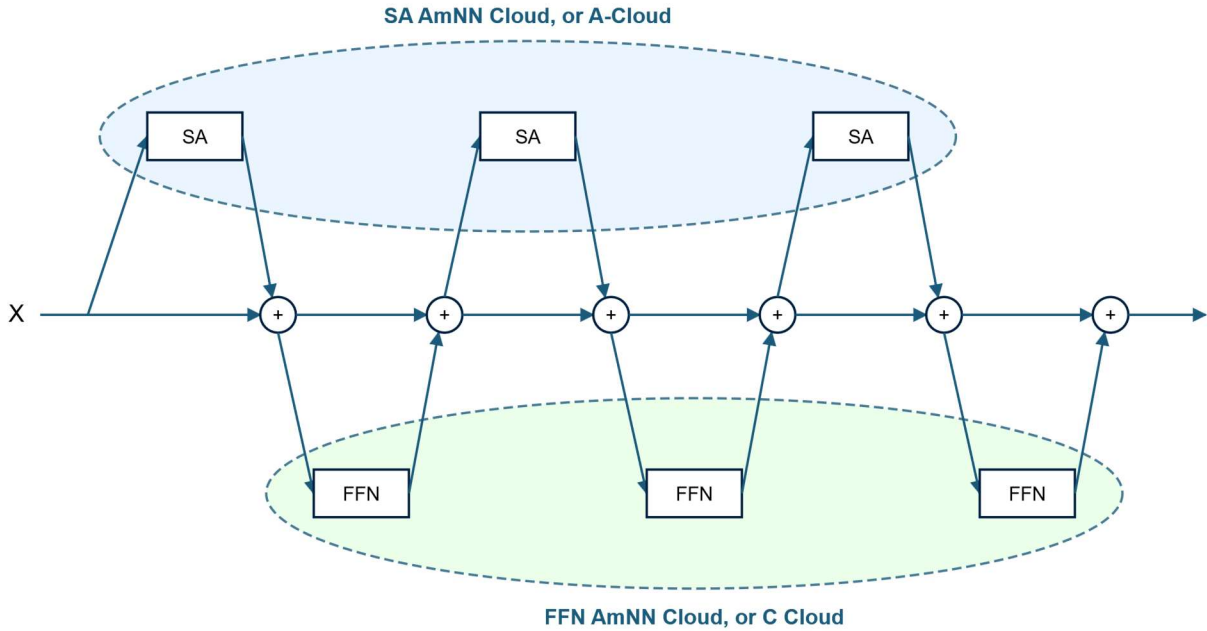


Figure 1: AmNN Cloud Overlay on Discrete Transformer Flow

The A-Cloud replaced all the self-attention operators, and the C-Cloud replaced all the FFN operators. Each cloud started as an unstructured continuous latent space. To limit parameter expansion and simulate a Directed Acyclic Graph (DAG), both clouds utilized continuous recurrent loops. Stochastic binary and temporal masks allowed sparse local and depth connections. The A-Cloud received an injection of the residual signal, processed it through a series of fluidic states, ran it through a linear projection, applied the causal mask and query/key/value matrices and the softmax operation across the sequence dimension, and then extracted the result into the residual stream X . The C-Cloud processed flow through a series of fluidic states, extracted the result into the residual stream X , and received an injection of the residual signal. Both clouds repeated their operations to simulate transformer “layers”.

Both clouds were injected from the residual stream through linear projections with a non-linear activation (LeakyReLU). Layer Normalization was not used. The C-Cloud injection points past the first layer were to the same state space as the extraction point, after the extraction. The A-Cloud injection point was the state space following the extraction point. The extractions from both clouds utilized dense linear projections, without masking, to bridge the wide amorphous latent spaces back to the dimensionality of the residual stream. These projected signals were added directly to the residual signal X to complete the residual connection, without the application of activation functions or layer normalization at the exit boundary.

This structure created a workbench to test whether amorphous clouds could freely form to replace the functionality of the discrete SA and FFN operators. The workbench was configured with four injection and extraction points per cloud, representing an equivalent test structure of four transformer layers. The token dimension was set to 128, sequence length to 128, C-Cloud width to 256, and A-Cloud width to 128. This yielded approximately 1.4 million trainable parameters in the backbone cloud structure, or 7.8 million total trainable parameters when including the token embeddings. The baseline standard transformer was given 7 layers and an FFN width of 512 to equalize the nominal trainable parameter count.

3. Operation and Results of the Dual-Cloud Architecture

The dual-cloud and the 7-layer standard transformers were trained on the WikiText-103 dataset for 50 epochs across 10 runs each, using identical seed sets for both architectures. The lowest validation loss points for each run were recorded, and the 10-run average minimum was computed. The standard 7-layer transformer achieved an average minimum validation loss of 4.5189. The 4-layer dual-cloud transformer achieved an average minimum validation loss of 4.5468. Given that 14% of the dual-cloud backbone parameters were inactive due to masking and reliance on a single attention head, the dual-cloud transformer demonstrated competitive performance. This empirical test validates the use of AmNN clouds as substitutes for discrete neural network operators within an interconnected system.

During data capture for the dual-cloud transformer, mean absolute weight magnitudes

$$Ratio = \frac{\text{mean}(|W_{upstream}|)}{\text{mean}(|W_{injection}|)}$$

were recorded at two key boundaries per layer: the entry intersection (comparing new cloud injection to continuous upstream flow) and the exit boundary (comparing internal cloud flow to the final extraction). The most significant diagnostic data emerged from the entry intersection. The following figure displays the Upstream-to-Injection ratios from the first three random seeds at Epoch 50:

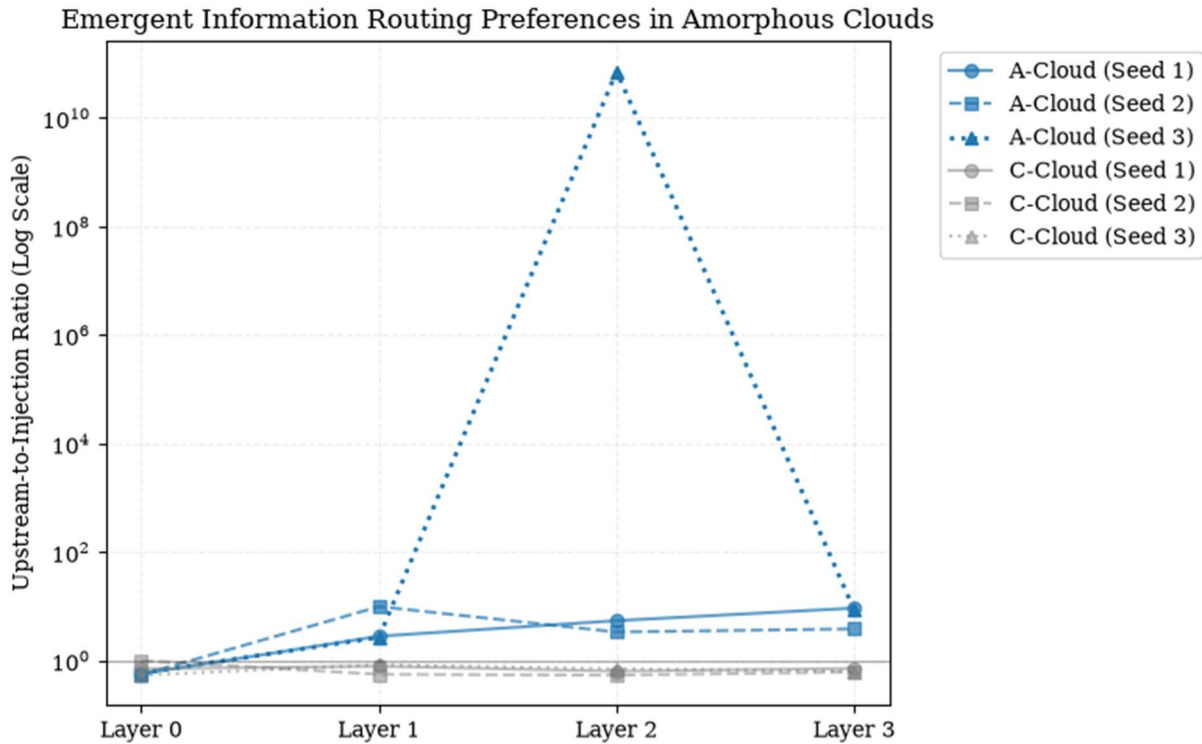


Figure 2: *Upstream-to-Injection weight ratios for the Amorphous A-Cloud and C-Cloud across three random seeds at Epoch 50. The A-Cloud demonstrates a consistent, learned preference for upstream latent context over residual stream injection; (Seed 3) completely severed the residual stream injection.*

In each of these examples, beginning at Layer 1, the self-attention processing in the A-Cloud weighted the continuous upstream fluidic data significantly heavier than the standard layer-normalized injection from the residual stream, X . Notably, in the third run at Layer 2, the A-Cloud exhibited a ratio of over 72 billion to one—indicating that the network effectively pushed the injection weights to zero, completely severing flow from the residual stream at that depth. This did not cause model instability or NaNs; it was the learned structural optimization.

This illustrates the unconstrained amorphous cloud dynamically generating its own routing structure. Across all ten runs, this trend of strong upstream signal preference in the A-Cloud held true for at least two of the four layers. This empirically demonstrates that the self-attention mechanism naturally seeks unmitigated "memory" context from earlier latent states, confirming that forcing attention modules to operate strictly from a normalized residual stream constitutes a structural inefficiency. The consistent natural internal structuring during training shows a demand for latent upstream information in self-attention. The C Cloud also showed a desire for

upstream data, but not as a preference over injected data. In these tests, the AmNN clouds were a powerful diagnostic tool.

4. Building a Better Transformer

The primary requirement for translating the insights of the dual-cloud analysis into a practical transformer is that the modifications must be minimally invasive and compatible with highly optimized, modern hardware. Inserting a continuous, unconstrained amorphous structure directly into standard production pipelines is not a viable option for today's industry.

4.1 Memory Highways

Arriving at the final, data-guided transformer design was a two-step process. The initial approach (Model I) sought to directly replicate the dual continuous information “highways” observed in the A-Cloud and C-Cloud. Because the upstream flow in the amorphous clouds is sourced from direct projections of the residual signal, X , this first discrete model added two parallel, unnormalized memory streams. These streams were constructed from latent data projected directly from the residual signal, with one stream providing unmitigated context to the self-attention operators, and the other to the FFN operators. The topology of this dual-highway architecture is illustrated in Figure 3.

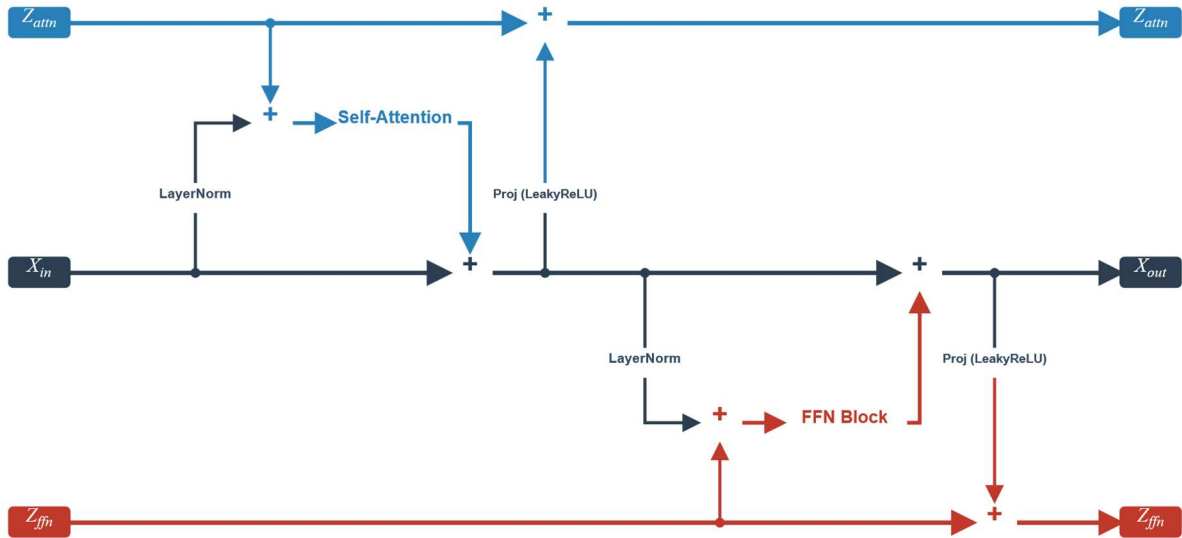


Figure 3:Transformer with dual memory highways

Compared to a standard transformer, this architecture introduces a minor per-layer parameter overhead to account for the linear projections into the memory highways, alongside an increased VRAM requirement to maintain the dual highway states in memory. To enforce strict parameter equalization with the baseline, this dual-highway transformer was constrained to 6 layers. This configuration contained approximately 1.4 million core backbone parameters and 7.8 million total trainable parameters.

The average minimum validation loss reached across 10 runs of 50 epochs each was 4.4438. Despite operating with one fewer layer, the 6-layer dual-highway transformer demonstrated a performance advantage over the 7-layer standard transformer baseline, which recorded a 10-run average minimum validation loss of 4.5189.

4.2 The Unified Memory Highway

Looking again at Figure 3, the two memory highways are mathematically similar; the difference is the values of the parameters in each of the memory projections from the spine. So a single memory highway Z_{mem} was implemented to feed both the self-attention and FFN blocks. In the first version, memory projections from the residual signal path after each of the discrete operators were added to Z_{mem} .

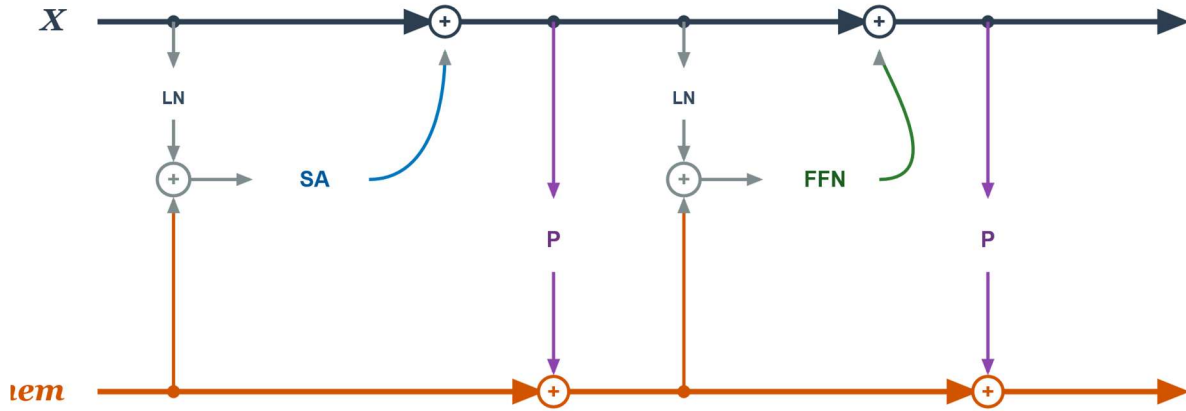


Figure 4: Unified Highway Transformer Layer

As the two memory streams did in the prior example, the unified memory stream acts as a stateful buffer that preserves unmitigated historical context, directly addressing the information degradation caused by layer normalization.

The Self-Attention Phase

Before executing the Self-Attention (SA) operation, the normalized primary stream is aggregated with the current state of the memory stream. This provides the discrete operator with direct access to the preserved upstream flow:

$$\begin{aligned} H_{SA}^{(l)} &= \text{LN}(X^{(l)}) + Z_{mem}^{(l)} \\ X_{mid}^{(l)} &= X^{(l)} + \text{SA}(H_{SA}^{(l)}) \end{aligned}$$

Following the SA block, the network executes the first memory projection P to the unified highway. The memory projection P extracts latent states from the updated residual stream and strictly adds them into the memory stream. This function P is parameterized as a linear projection followed by a LeakyReLU activation, allowing the network to selectively transform and filter the extracted features before integration:

$$Z_{mem,mid}^{(l)} = Z_{mem}^{(l)} + \text{LeakyReLU}\left(P_{SA}(X_{mid}^{(l)})\right)$$

The Feed-Forward Phase

This aggregated context is then carried forward into the Feed-Forward Network (FFN) phase. The FFN operator draws its input from the fusion of the newly normalized primary stream and the freshly updated memory stream:

$$\begin{aligned} H_{FFN}^{(l)} &= \text{LN}(X_{mid}^{(l)}) + Z_{mem,mid}^{(l)} \\ X^{(l+1)} &= X_{mid}^{(l)} + \text{FFN}(H_{FFN}^{(l)}) \end{aligned}$$

Finally, a second memory projection operation captures the semantic depth generated by the FFN, applying the same linear projection and LeakyReLU activation to inject the transformed state back into the memory stream before passing to the next layer:

$$Z_{mem}^{(l+1)} = Z_{mem,mid}^{(l)} + \text{LeakyReLU}\left(P_{FFN}(X^{(l+1)})\right)$$

Stateful Aggregation vs. Stateless Connections

This design fundamentally differs from standard residual skip connections.

- **Stateless Skips:** Standard residual connections are strictly additive pipelines that bypass discrete operators but are ultimately subjected to ablation via layer normalization at the next depth step.
- **Stateful Folding:** The memory stream acts as an independent, continuous accumulator. Because Z_{mem} bypasses layer normalization entirely, the memory projection operator P iteratively builds a representation of the sequence's history.

By replacing the rigid, serialized flow with this interconnected highway, the architecture provides discrete operators with the continuous context routing observed in the AmNN diagnostic models, significantly improving sample efficiency.

4.3 Testing and results of the Unified Highway

The Unified Highway Transformer (UHT) was evaluated across four progressively deep configurations. Test 1 evaluated a 6-layer UHT against a 7-layer parameter-equivalent standard Pre-LN transformer for 50 epochs; Test 2 evaluated a 17-layer UHT against a 20-layer parameter-equivalent baseline for 100 epochs; Test 3 evaluated an 80-layer UHT against an 80-layer standard transformer for 50 epochs; and Test 4 evaluated a 120-layer UHT against a 120-layer standard baseline. In all tests, identical weight initialization seeds were used across architectures to isolate topological impacts. Due to compute constraints, some comparative pairs were executed in parallel in separate hardware environments (e.g., a desktop-class GPU and cloud-based A100s/T4s). While the hardware variations may introduce minor floating-point variance, the magnitude of the performance divergence easily surpasses any performance variance attributable to hardware. Across all four scales, the Unified Highway achieved a lower validation loss than the standard baseline.

Shallow and Moderate Depths (Tests 1 & 2)

In Test 1 (10 runs each), the UHT reached an average validation loss of 4.43, compared to 4.52 for the standard transformer. At this depth, the Z_{mem} highway was updated with projections from the residual signal without any scaling, normalization, or clamping. While this theoretically raised the risk of exploding variance, none was observed at this shallow scale.

In Test 2 (3 runs each), an ablation run utilized a static scaler on the memory projections to address early signs of variance expansion. The UHT reached an average validation loss of 4.00, compared to 4.05 for the baseline. In the final configuration, the memory projections were strided (every other layer) and a Dynamic Variance Scaler was introduced to scale the memory projections for the first few epochs, progressing to a 1.0 scaling factor. The resulting average low validation loss was 3.89.

The 80-Layer Scaling Test and Dynamic Variance Annealing (Test 3)

Given the variance observed in Test 2, Test 3 (3 runs each) was designed as a strict stress test of scalability. To allow rapid iteration, the token dimension was bottlenecked to $d = 64$. This intentionally forced the learning into a highly compressed vector space, which naturally exacerbates variance expansion.

Initial attempts to train the 80-layer UHT revealed that static scaling was insufficient to control early-epoch variance. To solve this, we introduced a Dynamic Variance Scaler that progressively annealed the scale factor for the memory projections over the first 10 epochs. This created a highly stable Z_{mem} signal across all three runs, despite utilizing no normalization and no clamping.

In Test 3, the stabilized UHT reached an average low validation loss of 4.49, compared to an average low of 5.06 for the 80-layer standard transformer. Because this test was conducted at Iso-Layer rather than Iso-Parameter depth, the UHT carried a 5.6% total trainable parameter (7.5% backbone) overhead.

However, more significant than the validation loss differential is the time-to-capability acceleration. In Test 3, the lowest validation loss achieved by the standard baseline was 5.07 at Epoch 47. The UHT surpassed this capability limit at Epoch 13 (Runs 1 and 2) and Epoch 18 (Run 3). Taking the absolute worst-case scenario (Epoch 18) and penalizing for the 7.5% backbone overhead, the Unified Highway demonstrates a net Time-to-Capability acceleration of 59%. While this exact percentage is specific to this bottlenecked test, it empirically emphasizes that standard deep transformers are severely handicapped early in training due to PreNorm Dilution, and that resolving this bottleneck yields significant optimization efficiencies.

Extreme Depth and Latent Amnesia (Test 4)

Test 4 was designed as a deliberate architectural stress test to definitively isolate the impact of the un-normalized Z_{mem} highway at extreme depths. A 120-layer UHT was evaluated against a 120-layer standard Pre-LN Transformer over 50 epochs. To force representational compression and expose routing bottlenecks, both models utilized a constrained embedding dimension ($d_{model} = 64$) and identical hyperparameter configurations. To ensure rigorous comparability while allowing the standard Pre-LN transformer to run on a T4 processor, the baseline utilized gradient accumulation (two forward passes of batch size 8) to mathematically match the UHT’s native batch size of 16. The UHT was configured with a 2-stage strided topology (updating the memory highway strictly on alternating layers to halve projection friction) and a dynamic scale-free annealer. To absorb initialization shocks, this annealer began with an aggressive residual scaling divisor of $1/\sqrt{8L}$. Over the first 5 epochs, both the scalar multiplier and the layer-count dependency (L) were dynamically decayed until the divisor reached exactly 1.0. This completely removed all mathematical depth penalties, allowing full memory signals to accumulate on the memory highway, and forcing the network to natively self-regulate its variance for the remainder of training. Due to latent amnesia, the standard Pre-LN transformer would be expected to perform worse than the 80 layer standard Pre-LN transformer in Test 3. To show that the memory highway efficiently addresses latent amnesia at extreme depth, the 120 layer UHT would need to outperform the 80 layer UHT.

Initialization and the Illusion of Convergence:

The standard 120-layer Pre-LN baseline successfully survived initialization without experiencing numerical detonation (NaNs). During the first two epochs, the baseline demonstrated steady convergence, tracking closely with the UHT. This initial descent is likely driven entirely by the network’s shallowest layers, effectively masking gradient stagnation in deeper layers.

Observation of Latent Amnesia and Structural Uncoupling:

Monitoring the gradient decile profiles across all 120 layers revealed patterns associated with the rapid onset of Latent Amnesia in the baseline architecture. The standard Pre-LN model exhibited severe structural uncoupling. As the error signal was forced through 120 sequential normalization layers, the top 84 layers (closest to the loss function) became mathematically starved, with gradient magnitudes flatlining near ~ 0.05 . Conversely, the bottom-most layers absorbed compounding residual shocks, continuously spiking to magnitudes > 0.25 . This "pooling" effect indicates that deep Pre-LN layers become mathematically muted, forcing the network to operate with the effective capacity of a much shallower model.

Dynamic Routing and Scale-Free Stability:

In contrast, the strided UHT bypassed the sequential dilution of LayerNorm efficiently. Despite the complete removal of the depth divisor at Epoch 5, the UHT inherently prevented variance explosion, stabilizing its un-normalized memory highway variance at ~ 10.2 for the last 40 epochs. More importantly, the $O(1)$ routing of the Z_{mem} highway produced a healthy, non-monotonic "wave-like" gradient flow. Instead of pooling at the bottom of the stack, the UHT dynamically distributed compute, maintaining a strong foundational anchor at Layer 0 while spontaneously forming highly active, deep representational hubs (such as sustained gradient spikes at Layer 48).

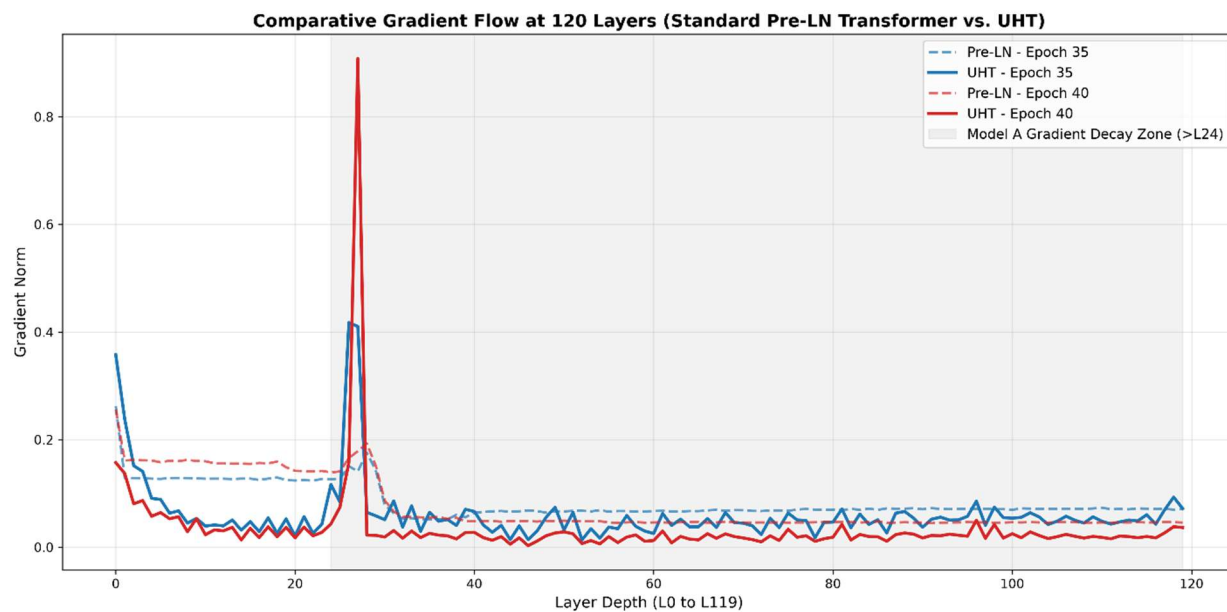


Figure 5: Gradient activity through 120 layers. The standard Pre-LN Transformer shows gradient stagnation in deep layers that is overcome in the UHT.

Pushing the Extreme:

To verify the absolute capacity limits of the 64-dimensional bottleneck, and to exercise the UHT at further depth, the UHT was configured with 180-Layers for a single test run. As shown in the following figure, this UHT also showed dynamic gradients at deep layers and verified the stability provided by the Dynamic Variance Annealer.

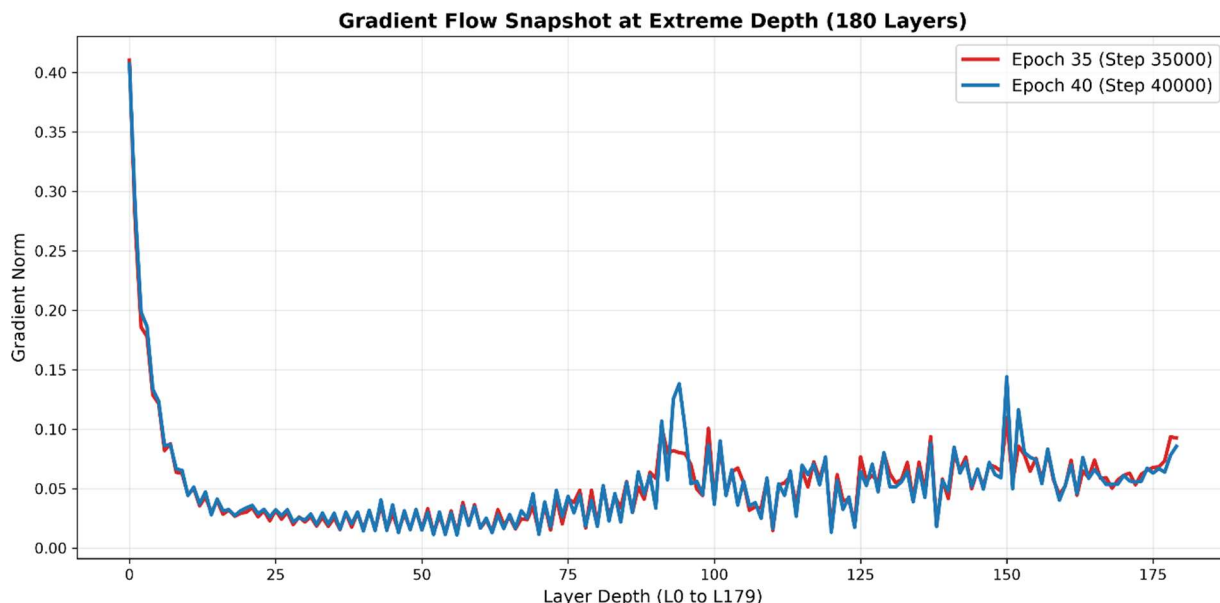


Figure 6: UHT Gradient activity through 180 layers; deep layer gradient stagnation is not observed.

Results:

The baseline standard transformer prematurely slowed its ability to learn complex semantic dependencies, and achieved a best validation loss of 5.1346 (Perplexity: 169.80) during the 50 epoch run. Despite having more parameters than the 80 layer Pre-LN Transformer, the 120 layer Pre-LN transformer did not match the 80 layer transformer’s performance. In contrast, the UHT preserved unimpeded gradient flow and actively utilized its full 120-layer depth, easily bypassing the Pre-LN’s performance and outperforming the 80 layer UHT, reaching an average low validation loss of 4.31 (Perplexity: 74). Showing continued improvement at depths that choke a standard transformer, the UHT empirically validates the strided Z_{mem} highway as a highly effective mechanism for overcoming Latent Amnesia at extreme depths.

The 180-layer UHT showed similar results as the 120-layer test, reaching a low validation loss of 4.28. This evidences that the model is reaching the performance limit of a $d_{model} = 64$ configuration.

5. Conclusion and Future Work

This work demonstrates that the rigid, serialized information flow of standard discrete-operator deep learning systems can be fundamentally improved by adopting continuous routing principles derived from Amorphous Neural Network (AmNN) diagnostic clouds. When substituted into a Pre-LN Transformer, the AmNN clouds demonstrated demand for upstream latent states. This work reinforced the theory that sequential Layer Normalization scrubs early-layer representations from the residual stream, ultimately leading to gradient stagnation.

This work led to the engineering of the UHT, which introduces the $O(1)$ parallel memory bus that accumulates and routes latent context directly to the deepest layers. When stabilized via Dynamic Variance Annealing and evaluated in deliberately narrow 120-layer and 180-layer stress tests, the Unified Highway successfully eliminated gradient stagnation that paralyzed a 120-layer Pre-LN baseline. By ensuring that structural depth translates directly into representational capacity, the Unified Highway Transformer continues to converge while the standard architecture predictably stalls.

Future Work

Having empirically validated the UHT's ability to improve Transformer performance, and to cure gradient stagnation at extreme depths in a constrained diagnostic setting, future work will investigate the integration of the UHT design across varying transformer architectures. Further work will explore the generalizability of this $O(1)$ memory bus to modalities beyond language modeling, while continuing to optimize the architecture for even greater parameter and inference efficiency. Ultimately, this research proves that Amorphous Networks are not merely theoretical novelties. They are invaluable analytical tools capable of diagnosing hidden architectural bottlenecks and guiding the design of highly effective, hardware-sympathetic improvements for industrial AI systems.

References:

Simon, A. L. (2026). *Amorphous Neural Networks*. U.S. Patent Application Publication No. US 2026/0252856 A1. Published August 27, 2026.

Kimi Team, Guangyu Chen, Yu Zhang, Jianlin Su, Weixin Xu, Siyuan Pan, Yaoyu Wang, et al. "Attention Residuals." *arXiv preprint arXiv:2603.15031* (2026).
<https://doi.org/10.48550/arXiv.2603.15031>.

Manson, Rob. "What Makes LLMs So Fragile (and Brilliant)?" *Flux by Rob Manson* (blog), December 1, 2025. <https://flux.robman.fyi/p/what-makes-llms-so-fragile-and-brilliant>.

AI Disclosure: An LLM (Gemini Pro) was utilized as a research assistant to generate initial Python code for the experiment baseline under the explicit structural and algorithmic specifications provided by the authors. All generated code was manually reviewed, verified, and debugged by the authors to ensure scientific and mathematical accuracy. The LLM was not allowed access to the code development environment or the document drafting environment. The LLM was also used as a writing assistant to suggest alternative phrasing for selected sections and to assist with general text cleanup and readability improvements. Every AI-generated suggestion was thoroughly reviewed, critically evaluated, and completely rewritten by the human authors. The final content, arguments, and claims presented in this work remain entirely the original contribution of the authors, who accept full responsibility for its integrity and accuracy.

Competing Interests: Aspects of the Unified Highway Transformer architecture, test methodology, and Amorphous Neural Networks are the subject of one or more pending patent applications filed by the author.